iTAML: An Incremental Task-Agnostic Meta-learning Approach

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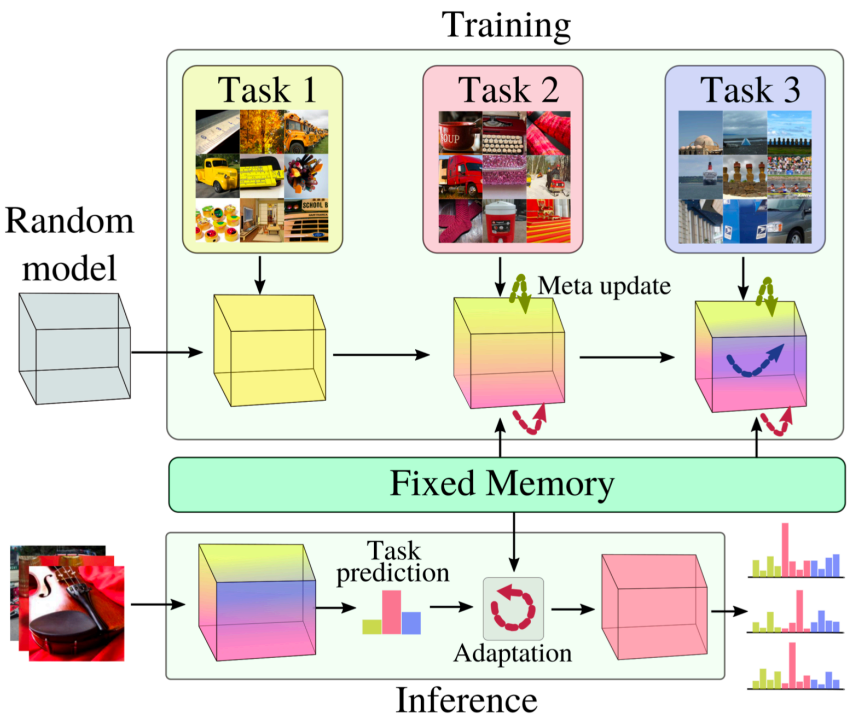
Incremental Task-Agnostic Meta-learning



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Incremental Task-Agnostic Meta-learning

- Generalization is a key for continual learning algorithms.
- Meta-learning achieves generalization by “learning to learn”.
- Adapting meta-learning for incremental learning involves many challenges.

Incremental Task-Agnostic Meta-learning

- SOTA incremental learning algorithms lack generalization.
- Meta-learning suffers on incremental learning setting.
    - Out of Order Distribution (OOD).
    - Often requires fine-tuning at the end.
    - Skewed data distribution with memory.
- iTAML tries to bridge the gap between meta-learning and incremental learning.

Incremental Task-Agnostic Meta-learning

- iTAML bridges this gap by:
  - A task-agnostic prediction mechanism, with two stage classification.
  - Disentangle the network into a generic feature extractor and task-specific classification weights.
  - A momentum based meta-update rule to avoid forgetting.
  - A sampling rate selection approach for data continuum.

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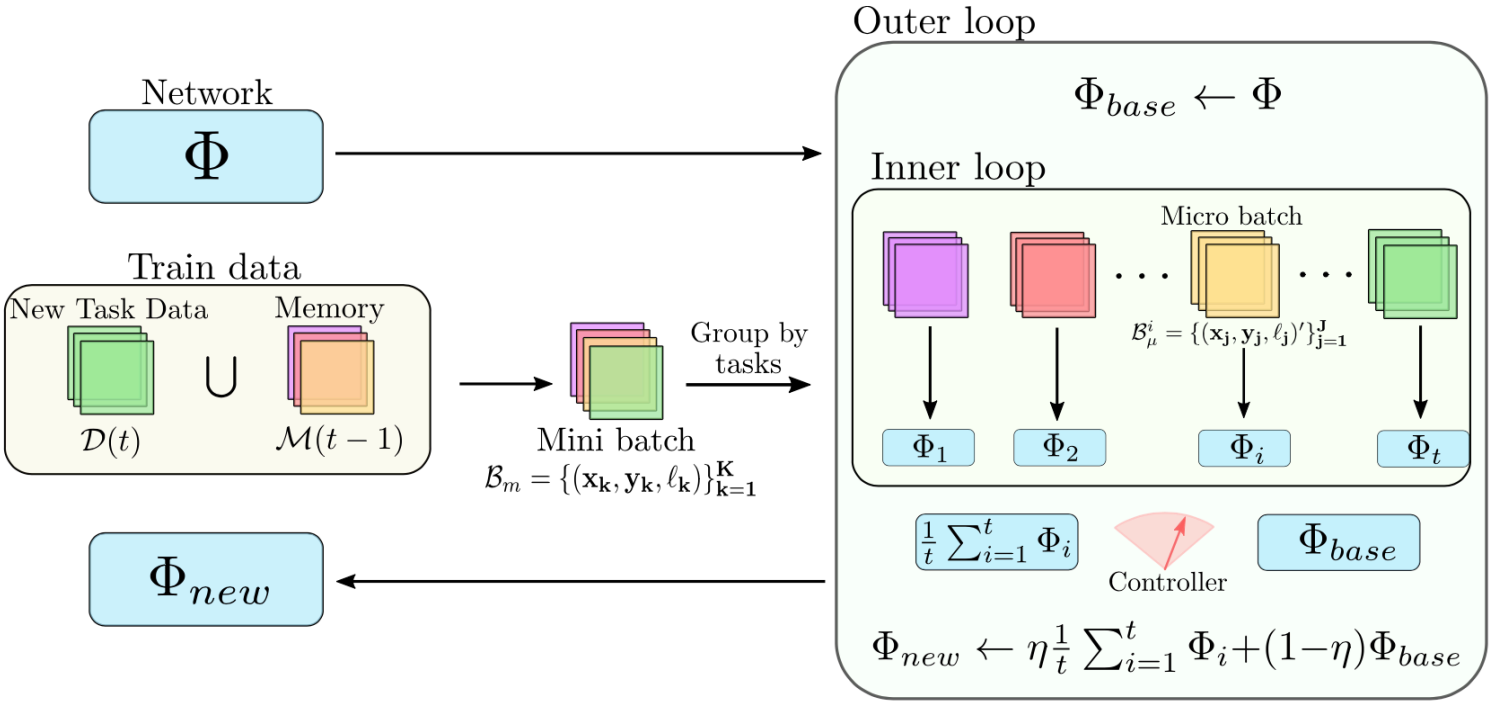
Meta Training of iTAML



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Meta Training of iTAML

- Each mini batch is further broken into task specific micro batches.
- In the inner loop, task specific models Φ_i are trained.
- In the outer loop a momentum controller combines these task specific weights.

$$\Phi_{new} = \eta \frac{1}{t} \sum_{i=1}^{t} \Phi_i + (1-\eta)\Phi_{base}.$$

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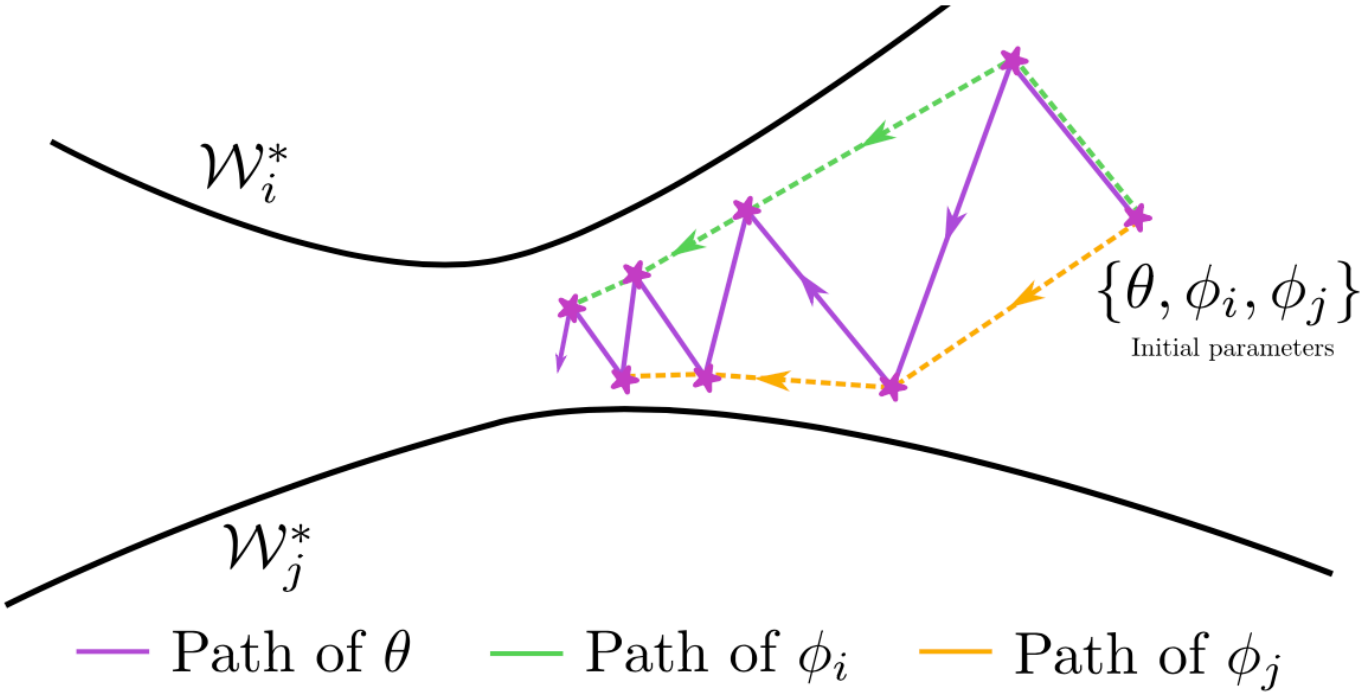
Meta Training of iTAML



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Meta Training of iTAML

- Since, the feature space parameters and classification parameters are tuned separately, iTAML remains task-agnostic.

- Feature space parameters, are tuned for each task and combined in the outer loop, hence they remain close to optimal solution manifold of all the tasks.

- Classification parameters are tuned only for the specific task; hence they remain close to the corresponding task's optimal solution manifold

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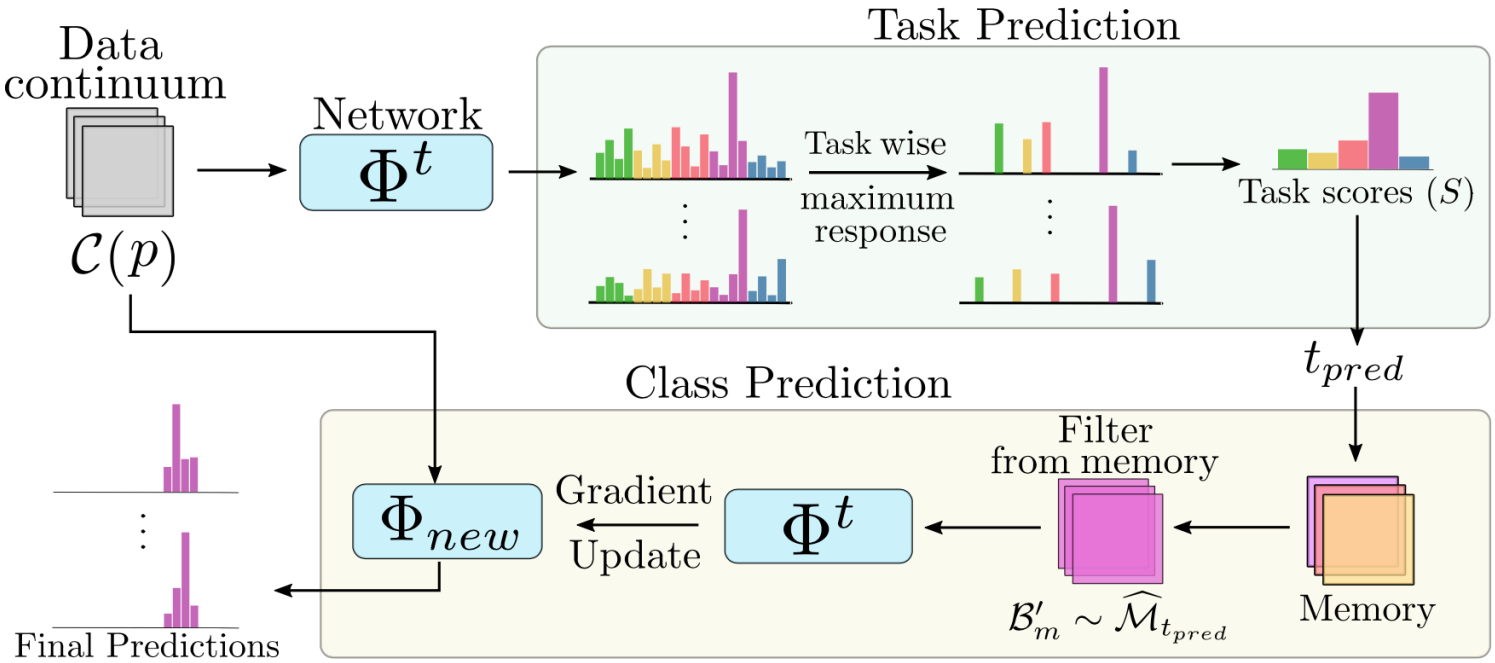
Inference of iTAML



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Inference of iTAML

- At inference, iTAML uses a two-stage prediction.
- First, given a data continuum $\mathcal{C}(i, p)$, it predicts the task using average predictions.
- Then, it uses exemplar data to adapt for the task using a gradient update.
- Finally, it passes the continuum and gives class-wise predictions.

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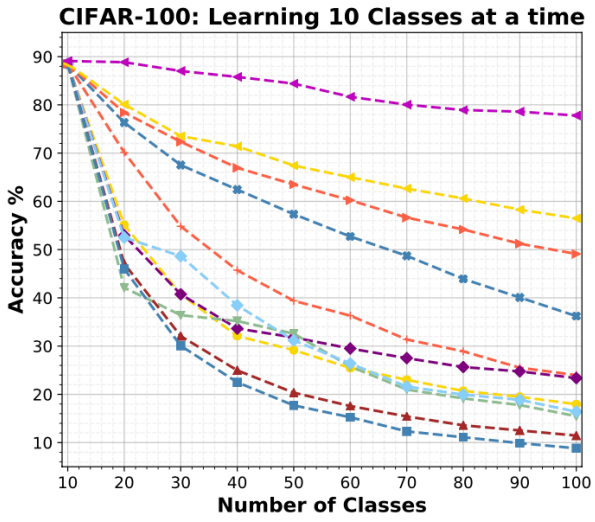
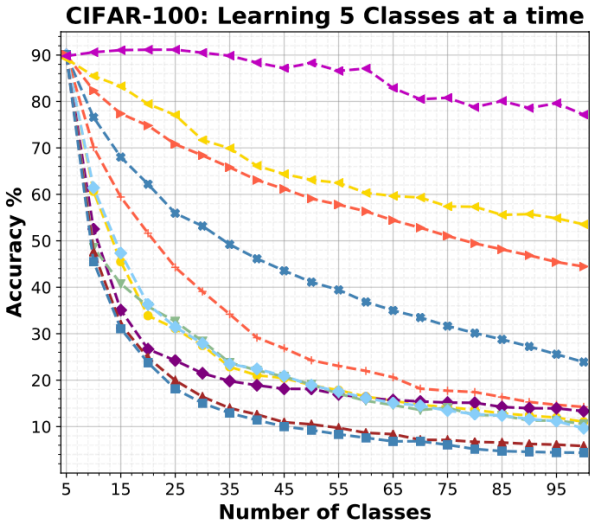
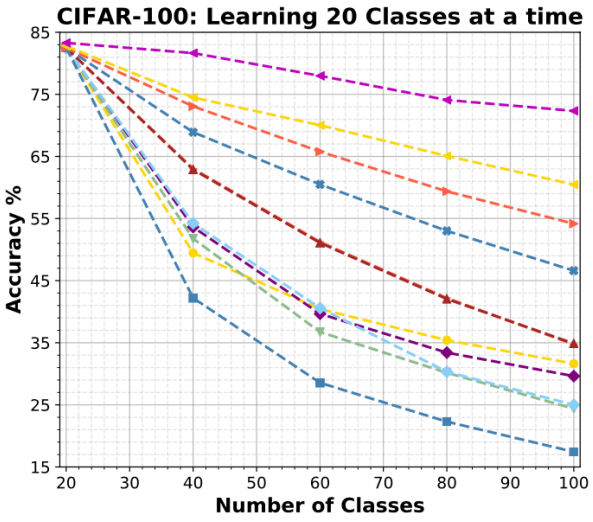
Few experimental Results







DMC LwF RWalk SI MAS EWC Finetuning FixedRep iCaRL RPS Ours

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Few experimental Results

Datasets	Methods	1	2	3	4	5	6	7	8	9	Final
ImageNet-100/10	Finetuning	99.3	49.4	32.6	24.7	20.0	16.7	13.9	12.3	11.1	9.9
	FixedRep	99.3	88.1	73.7	62.6	55.7	50.2	42.9	41.3	39.2	35.3
	LwF(TPAMI'18)[15]	99.3	95.2	85.9	73.9	63.7	54.8	50.1	44.5	40.7	36.7
	iCaRL(CVPR'17)[23]	99.3	97.2	93.5	91.0	87.5	82.1	77.1	72.8	67.1	63.5
	RPSnet(NeurIPS'19)[22]	100.0	97.4	94.3	92.7	89.4	86.6	83.9	82.4	79.4	74.1
	Ours	99.4	96.4	94.4	93.0	92.4	90.6	89.9	90.3	90.3	**89.8** $_{+15.7}$
ImageNet-1K/10	Finetuning	90.2	43.1	27.9	18.9	15.6	14.0	11.7	10.0	8.9	8.2
	FixedRep	90.1	76.1	66.9	58.8	52.9	48.9	46.1	43.1	41.2	38.5
	LwF(TPAMI'18)[15]	90.2	77.6	63.6	51.6	42.8	35.5	31.5	28.4	26.1	24.2
	iCaRL(CVPR'17)[23]	90.1	82.8	76.1	69.8	63.3	57.2	53.5	49.8	46.7	44.1
	Ours	91.5	89.0	85.7	84.0	80.1	76.7	70.2	71.0	67.9	**63.2** $_{+19.1}$
MS-Celeb-10K/10	iCaRL(CVPR'17)[23]	94.2	93.7	90.8	86.5	80.8	77.2	74.9	71.1	68.5	65.5
	RPSnet(NeurIPS'19)[22]	92.8	92.0	92.3	90.8	86.3	83.6	80.0	76.4	71.8	65.0
	BiC(CVPR'19)[30]	95.7	96.5	96.5	95.7	95.1	94.2	93.2	91.7	90.0	87.6
	Ours	94.0	95.6	96.0	95.8	95.5	95.4	95.2	95.1	95.0	**95.0** $_{+7.4}$

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Thank You!

https://github.com/brjathu/iTAML